



Climate Change and Plant Diseases: Altering the Global Path system



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Abstract

Climate change is rapidly altering the biological and environmental conditions that shape crop health, placing plant diseases at the forefront of global food security challenges. This abstract explores how rising temperatures, shifting precipitation patterns, and increasing atmospheric CO₂ interact with the disease triangle host, pathogen, and environment. Warming trends drive the expansion of pathogens and vectors into new regions, weaken host resistance, and modify pathogen aggressiveness. At the same time, moisture extremes intensify disease outcomes, favouring flood-associated pathogens and increasing susceptibility to drought-related diseases. These changes complicate disease prediction and control, underscoring the need for climate-resilient crop varieties and adaptive, flexible integrated pest management strategies.

Introduction

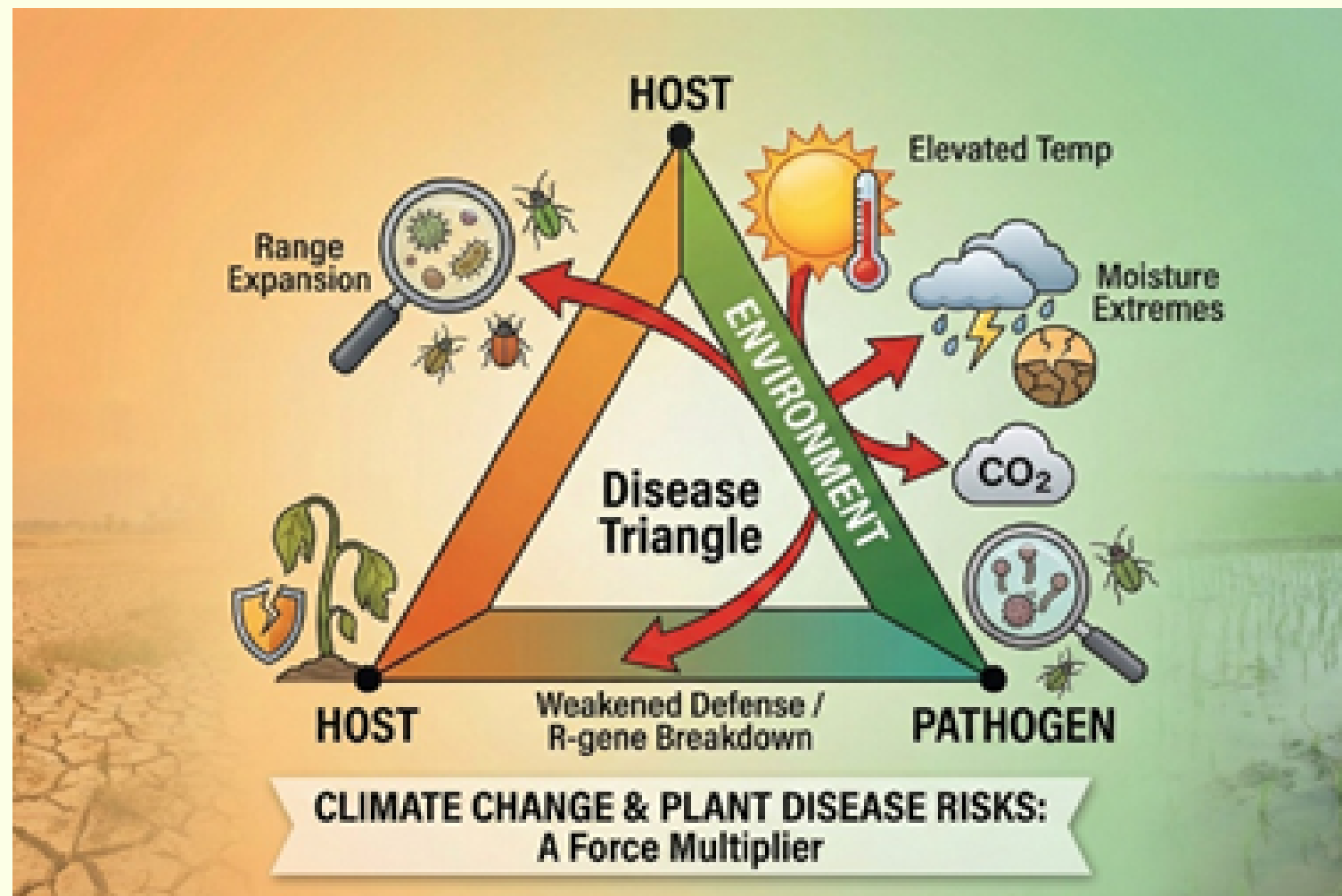
Plant pathology has long relied on the "Disease Triangle" concept, that disease emerges only when a susceptible host, an aggressive pathogen, and a favourable environment collide (Fig. 1). For years, the environmental corner of this triangle felt dependable. Seasonal. Largely predictable. That assumption no longer holds. Human-driven climate change is destabilising this balance at an unprecedented pace. Global surface temperatures are expected to rise by 1.5 - 2 °C over the course of this century, while rainfall patterns become increasingly erratic and atmospheric CO₂ levels, now above 415 ppm continue to increase. These forces do not act in isolation. They reshape plant physiology, accelerate or disrupt pathogen life cycles, and alter vector behaviour in subtle but consequential ways. The results are already unfolding in farmers' fields: unfamiliar disease complexes, once-minor pathogens gaining prominence, and tropical diseases advancing into temperate regions. Grasping these shifts is no longer an academic exercise it has become central to safeguarding global crop production (Williams & Boehm, 2017; Ziska, 2025).

Climate-Driven Modifications in Plant–Pathogen Interactions

Elevated Temperature and Pathogen Range Expansion

Temperature acts as a central regulator of pathogen metabolism, and rising global temperatures are reshaping disease behaviour in multiple, interconnected ways. Warmer winters now allow pathogens once constrained by cold to persist and spread into higher latitudes, as seen in wheat rusts (*Puccinia spp.*), which increasingly overwinter in regions where they previously failed to survive, triggering earlier and more intense spring epidemics. Elevated temperatures also shorten pathogen latency periods, enabling fungi and bacteria to complete life cycles more rapidly and generate multiple infection cycles within a single season (polycyclic diseases), sharply increasing inoculum pressure. At the same time, a milder winter conditions enhance the survival of pathogen propagules (sclerotia, spores) and insect vectors, ensuring a higher initial inoculum at the start of subsequent cropping seasons and intensifying disease outbreaks.

Fig.1 Depicts climate change as a force multiplier that destabilizes host-pathogen-environment interactions, driving severe plant disease outbreaks.



Contrasting Effects of Flooding and Drought on Plant Health

Climate change increasingly expresses itself as hydrologic volatility, marked by sharp swings between intense rainfall and prolonged drought, each favouring a different set of plant diseases. Periods of excess moisture create ideal conditions for water-dependent pathogens, particularly oomycetes such as *Phytophthora* spp. and *Pythium* spp., where saturated soils enhance zoospore movement and drive severe root rots and damping-off. Elevated humidity within dense canopies further accelerates foliar diseases, including blights and downy mildews. In contrast, extended drought imposes physiological stress on plants, weakening their defence capacity and increasing vulnerability to necrotrophic and opportunistic pathogens such as *Macrophomina* (charcoal rot) and *Fusarium* (Vascular wilt). Reduced plant water potential under drought conditions limits the mobilisation of defensive compounds, tipping the balance in favour of pathogen invasion and disease development (MacDonald, 1982; Devi et al., 2017).

Influence of Elevated Atmospheric CO₂ on Plant-Pathogen Interactions

Rising atmospheric CO₂ generally enhances growth and biomass accumulation in C₃ crops; however, this apparent benefit comes with important disease-related trade-offs. Increased vegetative growth leads to denser crop canopies that trap moisture and restrict air movement, creating a favourable microclimate for fungal sporulation and bacterial proliferation, as seen in diseases such as *Sclerotinia* white mould. Elevated CO₂ also alters plant tissue composition, typically reducing nitrogen content while increasing carbon-to-nitrogen ratios, which can slow the growth of some biotrophic pathogens yet promote greater cellulose and lignin deposition, influencing residue decomposition and the survival of saprophytic pathogens. At the same time, certain fungi, including *Colletotrichum* species, exhibit enhanced aggressiveness under high CO₂, producing additional virulence factors to breach strengthened host structural defences (Koo et al., 2016).

Vector Biology and Viral Spread

The influence of climate change on viral plant diseases is largely channelled through its effects on insect vectors such as aphids, whiteflies, thrips, and leafhoppers. Rising temperatures accelerate vector reproductive rates, increasing the number of generations produced each year and driving larger vector populations that intensify virus transmission. At the same time, warming conditions are enabling vectors to expand into higher latitudes and altitudes; the spread of the whitefly (*Bemisia tabaci*), for example, has facilitated the introduction of geminiviruses into regions that were previously unsuitable for vector survival. Climatic stress, particularly drought, also modifies vector behaviour, prompting more frequent movement and probing between plants, which enhances the efficiency of non-persistent virus transmission (Van Munster *et al.*, 2017).

Breakdown of Host Resistance

One of the most troubling consequences of rising temperatures is the growing instability of resistance (R) genes that underpin modern disease management. Many R genes deployed in breeding programs are inherently temperature sensitive, losing effectiveness under elevated thermal conditions. A well-known example is the Mi gene in tomato, which provides resistance against root-knot nematodes (*Meloidogyne spp.*) but fails when soil temperatures exceed approximately 28 °C. Similar temperature-driven breakdowns have been reported for resistance to Wheat streak mosaic virus. These patterns indicate that crop varieties classified as “resistant” may become vulnerable during heat waves, raising serious concerns that climate extremes could undermine decades of genetic improvement and complicate future breeding strategies (Devran *et al.*, 2023).

Conclusion and Future Perspectives: Towards Climate-Smart Pathology

Climate change has emerged as a powerful force multiplier for plant disease risk, destabilising the long-held balance of the disease triangle. Temperature shifts, erratic moisture regimes, and rising atmospheric gases have transformed the environment from a passive backdrop into an active driver of epidemics. These changes favour pathogen survival, speed up evolutionary processes, and weaken host defence systems. As a result, diseases once considered minor are gaining prominence, tropical pathogens are encroaching into temperate regions, and complex disease interactions such as nematode-fungal wilt complexes under stress are becoming more frequent. For producers, this translates into higher production costs from increased fungicide use, unexpected yield losses, and the erosion of established quarantine boundaries. Confronting this reality requires a decisive shift away from reactive, chemistry-heavy disease control. Erratic weather patterns increasingly undermine the effectiveness of routine spray schedules, making a proactive, systems-based approach essential. This transition is best captured by the concept of Climate-Smart Plant Pathology, which integrates ecological insight with technological innovation to anticipate and manage disease risk before outbreaks escalate.

At the core of this approach lies next-generation breeding. Disease resistance alone is no longer sufficient; breeding programs must prioritise climate resilience alongside resistance. Identifying and stacking heat-stable resistance genes that retain functionality under thermal stress is critical. Precision tools such as CRISPR-Cas9 offer opportunities to introduce broad-spectrum, durable resistance traits capable of withstanding fluctuating environmental conditions. Disease management strategies must also evolve. Conventional Integrated Pest Management calendars, built on historical climate norms, are rapidly losing relevance. In their place, dynamic, real-time IPM systems are needed. By combining remote sensing, weather data, and AI-driven predictive models, disease outbreaks can be forecast days in advance, enabling targeted preventive interventions such as timely biocontrol applications rather than blanket chemical spraying.

Adaptation cannot stop at the field edge. Cropping systems themselves must be redesigned to cope with accelerated pathogen life cycles. Diversified crop rotations help disrupt soil-borne pathogens like *Fusarium* and plant-parasitic nematodes that thrive in stressed monocultures. Improving soil health, particularly through increasing organic matter, strengthens beneficial microbial communities and promotes suppressive soils, offering a biological buffer against climate-induced disease susceptibility.

Finally, addressing the intersection of climate change and plant disease is a global imperative. Strengthened international cooperation is needed to track the poleward movement of pathogens and vectors through robust phytosanitary surveillance. Supportive policy frameworks must accelerate the approval of biopesticides and climate-resilient seed varieties. Protecting plant health in a warming world will depend on a holistic strategy, one that connects molecular biology, agronomy, and climate science to safeguard global food security in an increasingly volatile environment.

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